

Samir Orucov

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Summary

Holder of a PhD in applied mathematics, specializing in financial econometrics, **machine learning**, and **numerical optimization**. Within the framework of doctoral research, developed novel hybrid models such as **log-TGARCHX**, **Clustered HAR**, and **TAR-DT** for modeling the conditional volatility of the Bitcoin market; applied LASSO, ABESS, and Boruta methods for variable selection, and the Decision Tree machine learning algorithm for studying regime dynamics. Conducted parallel Monte Carlo simulations on multi-core CPUs to verify the statistical properties of algorithms. Currently serves as Assistant Professor in Mathematics and Statistics at ADA University and Head of the Statistics Division at the Information and Communication Technologies Agency. Possesses professional-level skills in Python and R programming languages, with extensive experience in time series analysis, big data analytics, and the development of automated statistical systems. Strives to contribute to the creation of innovative data-driven solutions in the telecommunications and finance sectors by combining academic research with practical application.

Skills

Machine Learning	Variable selection (LASSO, ABESS, Boruta), Decision Tree (CART), K-means clustering, Cross-Entropy optimization, ensemble methods (XGBoost, Random Forest), conditional volatility modeling
Parallel Computing	Parallel Monte Carlo simulations on multi-core CPU (Python multiprocessing/joblib), processing of high-volume financial data
Programming	Python (scikit-learn, pandas, NumPy, statsmodels, joblib), R (tidyverse, forecast, rugarch, gets), SQL, MATLAB, Stata
Econometrics	EViews, Stata, R (plm, AER), Python (statsmodels) — time series, panel data, IV methods, log-GARCH/HAR models
Development Environment	VS Code, Spyder, RStudio, LaTeX, Quarto, Git/GitHub, DBeaver
Other	Google Apps Script, Power BI, MySQL, Bash/PowerShell

Work Experience

- 03/2025– Present **Assistant Professor in Mathematics and Statistics, ADA University, Baku, Azerbaijan**
- **Teaching:** Mathematical Statistics, Applied Business Statistics, Econometrics, Microeconomics, Macroeconomics, Introduction to Higher Mathematics.
 - **Research:** Modeling and forecasting of financial time series using machine learning and deep learning methods (LSTM, GRU, XGBoost); AI-based estimation of conditional volatility in the Bitcoin market; analysis of high-frequency financial data using parallel computing architectures; application of extreme value theory in estimating the probability of rare events.

- 03/2025– **Head of Statistics Division (secondary employment)**, *Information and Communication Technologies Agency*, Baku, Azerbaijan
Present
- Building automated verification systems and anomaly detectors to improve the quality of statistical data.
 - Automated preparation of reports on the telecommunications market at various frequencies.
 - Programs for automated collection of data from websites.
- 06/2024– **Lead Specialist**, *Information and Communication Technologies Agency*, Baku, Azerbaijan
03/2025
- Programs and reports for real-time internet speed analysis using the Ookla database.
 - Automation of PDF report generation for data visualization.
- 01/2019– **Lecturer in Economics and Statistics**, *ADA University*, Baku, Azerbaijan
03/2025
- Micro/Macroeconomics (4 sections), Econometrics (Python/R integration), Business Statistics.

Education

- 12/2024 **PhD in Applied Mathematics**, *University of Southern Brittany*, Vannes, France
Full tuition waiver.
- Developed the CESBKM algorithm combining Cross-Entropy methods with K-means for clustering.
 - Conducted theoretical proofs and simulation studies demonstrating the superiority of the algorithm.
- 01/2019 **Master's in Economics**, *CERGE-EI (Center for Economic Research and Graduate Education – Economics Institute)*, New York, USA
Full CERGE-EI scholarship (full tuition waiver).
- Courses: Statistics, Econometrics I/II, Time Series Econometrics, Empirical Methods, Financial Econometrics.
 - Notable Achievement: Received an A- grade in the General Exam in Statistics/Econometrics.
- 01/2012 **Master's in Business and Regulatory Law**, *Free University of Berlin*, Berlin, Germany
Full scholarship of the Government of Azerbaijan (full tuition payment).
- 01/2009 **Bachelor's in Law**, *Baku State University*, Baku, Azerbaijan
Presidential Scholarship (by Decree No. 994 of the President of the Republic of Azerbaijan, dated September 12, 2005).

Publications

- 2025 "The Transformation of Broadband Demand: From Discretionary Service to Essential Infrastructure (2010–2024)," *SSRN Working Paper No. 5920805*, Orujov, S., Ismayilov, I., Huseynzade, J., 2025.
- 2024 "VS-LTGARCHX: A Flexible Variable Selection Algorithm for log-TGARCHX Models," *Journal of Time Series Econometrics*, De Gruyter, 2024.
- 2019 "Assessment of the state and classification criteria of small and medium enterprises in Azerbaijan," *Proceedings of ANAS (Economics series)*, 07/2019.
- 2019 "Econometric assessment of the impact of small and medium enterprises on economic development," *Journal of Information and Innovations*, Russia, 01/2019.

Languages

Azerbaijani	Native
English	Fluent
Russian	Fluent